

USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LAB #4 (Lesson 17) - Introduction to Phased Array Hardware
Lab Sheet
Fall 2026

Name: _____ **Section:** _____ **Date:** _____

Learning Objective 3.3: I can identify the hardware architecture of the ADALM-PHASER and control it via SDR software.

This sheet is the turn-in document for the Lesson 17 lab. Fill in every box, and show the arithmetic where a box asks for a calculated value. The lab is a team assignment, so each team submits one sheet.

- 1. The frequency plan.** Record the frequency the GUI reports for your HB100 after pressing **Find HB100**, then work the high-side mixing arithmetic that follows from it. The IF is fixed by the filtering after the mixer, so the LO is the only frequency that moves when the source moves.

Quantity	Value
Measured HB100 frequency f_{RF}	
Required LO frequency f_{LO}	
Resulting IF f_{IF}	
Is the LO inside 12.2–13.0 GHz?	

Show the arithmetic for your measured frequency here.

- 2. The Labeled Block Diagram.** Sketch the receive chain from the patches to the Raspberry Pi. Label every block with its name and with the frequency present at that point, mark where the analog summing happens, and mark where the two digital channels begin.



3. FFT observations. Work on the FFT tab with the source at 1 m and record what the spectrum does as you change two controls.

(a) Where does the tone sit on the FFT tab's frequency axis, and how far above the noise floor is it?

Peak offset =

Peak above floor =

(b) Set **Rx Gain (dB)** to 10 and record the peak and floor, then repeat at 0 and at 20. Report the peak and floor levels in dBFS (decibels relative to the ADC's full-scale input) and the separation between them in dB.

Rx Gain	Peak (dBFS)	Floor (dBFS)	Separation (dB)
0 dB			
10 dB			
20 dB			

(c) Lower **Signal Freq (GHz)** by 0.001 GHz (one click of the field's down arrow) and press Enter, then record how far the peak moves and in which direction. (On an older kit, press **Stop** and then **Start** to send the new value.)

Peak shift =

4. Written answers. Answer each in three or four sentences.

- (a) Why does the control software have to measure the HB100's frequency instead of assuming 10.525 GHz?

- (b) The array has eight elements, but software sees only two digital channels. Explain where the other six went, and name one measurement this rules out.

Documentation: _____